PROJECT 3

Active Directory & Windows Security Lab

Evidence-Based Technical Documentation

Domain **lab.local**	**Domain Controller** **DC01**
Workstation **CLIENT01**	**Network** **VMware NAT**

Built from the actual lab screenshots captured during Project 3.

1. Executive Summary

This project built and validated a Windows enterprise security environment using Windows Server 2022 and Windows 11. The lab established Active Directory and DNS, structured directory objects with OUs, joined CLIENT01 to the domain, applied workstation security policy, enabled advanced auditing, verified Windows Security events, enabled Windows Defender Firewall controls, and hardened the endpoint.

2. Lab Architecture

The environment consisted of DC01 running Windows Server 2022 with Active Directory Domain Services and DNS, and CLIENT01 running Windows 11 as a domain workstation. The virtual machines communicated through VMware NAT.

Validation: CLIENT01 successfully resolved lab.local to 192.168.136.131.

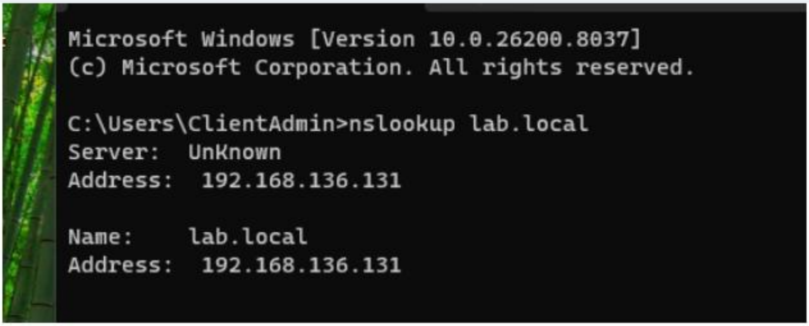


Figure 1 — CLIENT01 DNS validation: lab.local resolves to 192.168.136.131.

3. Active Directory Structure

The directory was organized into dedicated OUs: Admins, Security Groups, Servers, and Workstations. CLIENT01 was placed in Workstations so workstation-specific Group Policy could be targeted to it.

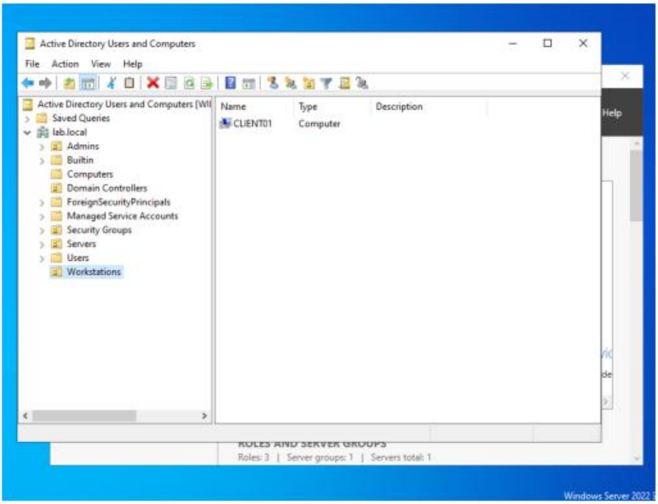
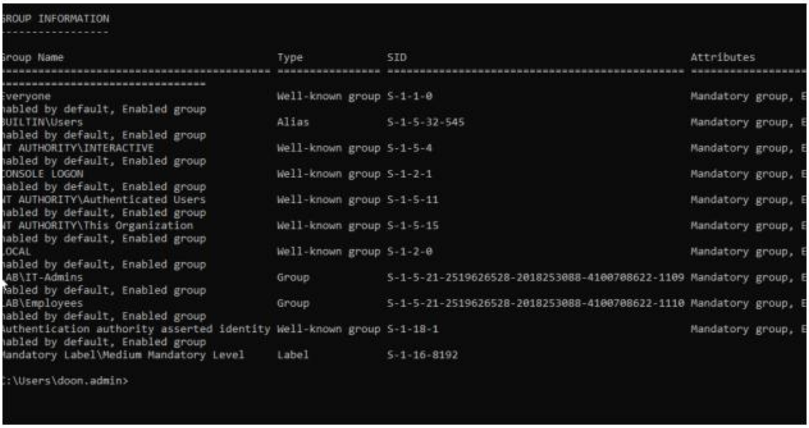


Figure 2 — Active Directory structure and Workstations OU.

4. Users & Security Groups

The lab created the domain account doon.admin and the security groups Employees and IT-Admins. The captured evidence shows the custom LAB security groups.



```
GROUP INFORMATION
-----------------

Group Name                                 Type             SID                                            Attributes
========================================== ================ ============================================== ==================
===============================
Everyone                                   Well-known group S-1-1-0                                        Mandatory group, E
nabled by default, Enabled group
BUILTIN\Users                              Alias            S-1-5-32-545                                   Mandatory group, E
nabled by default, Enabled group
NT AUTHORITY\INTERACTIVE                   Well-known group S-1-5-4                                        Mandatory group, E
nabled by default, Enabled group
CONSOLE LOGON                              Well-known group S-1-2-1                                        Mandatory group, E
nabled by default, Enabled group
NT AUTHORITY\Authenticated Users           Well-known group S-1-5-11                                       Mandatory group, E
nabled by default, Enabled group
NT AUTHORITY\This Organization             Well-known group S-1-5-15                                       Mandatory group, E
nabled by default, Enabled group
LOCAL                                      Well-known group S-1-2-0                                        Mandatory group, E
nabled by default, Enabled group
LAB\IT-Admins                              Group            S-1-5-21-2519626528-2018253088-4100708622-1109 Mandatory group, E
nabled by default, Enabled group
LAB\Employees                              Group            S-1-5-21-2519626528-2018253088-4100708622-1110 Mandatory group, E
nabled by default, Enabled group
Authentication authority asserted identity Well-known group S-1-18-1                                       Mandatory group, E
nabled by default, Enabled group
Mandatory Label\Medium Mandatory Level     Label            S-1-16-8192

C:\Users\doon.admin>
```

Figure 3 — Security groups in the LAB domain.

5. Workstation Security Baseline

A dedicated Group Policy Object named Workstation Security Baseline was created for the Workstations OU.

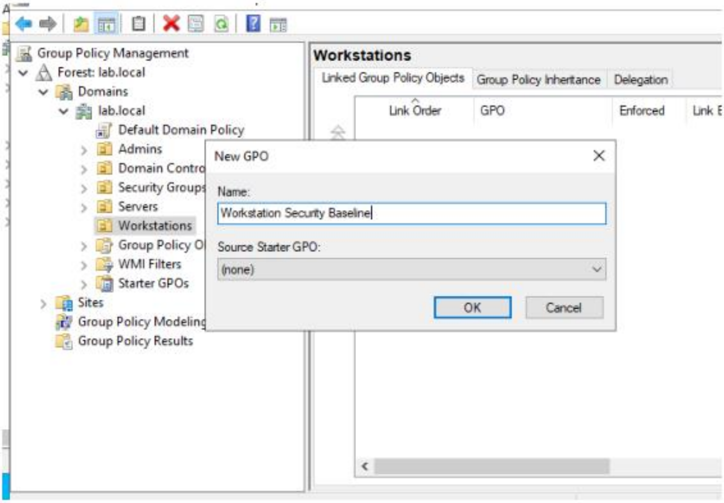


Figure 4 — Workstation Security Baseline GPO.

The policy was refreshed and then verified from CLIENT01. GPRESULT provides result-level evidence that the policy was actually applied rather than merely created.

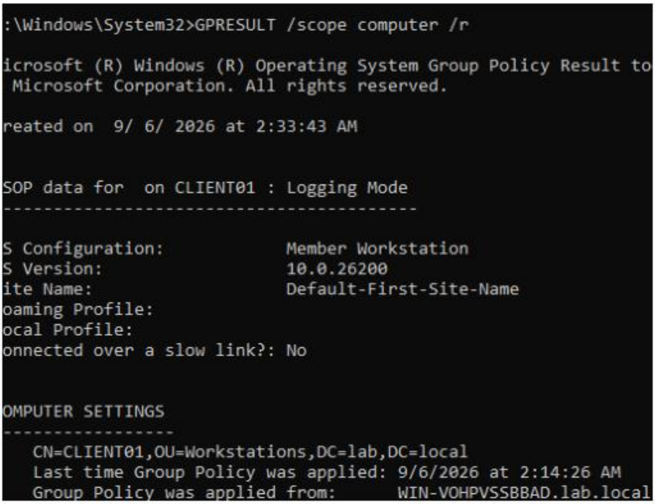


Figure 5 — GPRESULT validation on CLIENT01.

```
COMPUTER SETTINGS
------------------
    CN=CLIENT01,OU=Workstations,DC=lab,DC=local
    Last time Group Policy was applied: 9/6/2026 at 2:14:26 AM
    Group Policy was applied from:      WIN-VOHPVSSBBAD.lab.local
    Group Policy slow link threshold:   500 kbps
    Domain Name:                        LAB
    Domain Type:                        Windows 2008 or later

    Applied Group Policy Objects
    -----------------------------
        Workstation Security Baseline
        Default Domain Policy

    The following GPOs were not applied because they were filtered out
    -------------------------------------------------------------------
        Local Group Policy
            Filtering:  Not Applied (Empty)

    The computer is a part of the following security groups
    -------------------------------------------------------
        BUILTIN\Administrators
        Everyone
        BUILTIN\Users
        NT AUTHORITY\NETWORK
        NT AUTHORITY\Authenticated Users
```

Figure 6 — Applied Group Policy objects.

6. Advanced Security Auditing

Advanced auditing was configured for authentication, logon/logoff, account management, group management, and policy changes. The auditpol output provides direct configuration evidence.

```
  Plug and Play Events                    No Auditing
  Token Right Adjusted Events             No Auditing
Policy Change
  Audit Policy Change                     Success and Failure
  Authentication Policy Change            No Auditing
  Authorization Policy Change             No Auditing
  MPSSVC Rule-Level Policy Change         No Auditing
  Filtering Platform Policy Change        No Auditing
  Other Policy Change Events              No Auditing
Account Management
  Computer Account Management             No Auditing
  Security Group Management               Success and Failure
  Distribution Group Management           No Auditing
  Application Group Management            No Auditing
  Other Account Management Events         No Auditing
  User Account Management                 Success and Failure
DS Access
  Directory Service Access                No Auditing
```

Figure 7 — auditpol security auditing configuration.

7. Security Event Validation

Windows Security logs were reviewed to confirm that auditing generated actual events. Event ID 4634 records a logoff event associated with LAB\doon.admin.

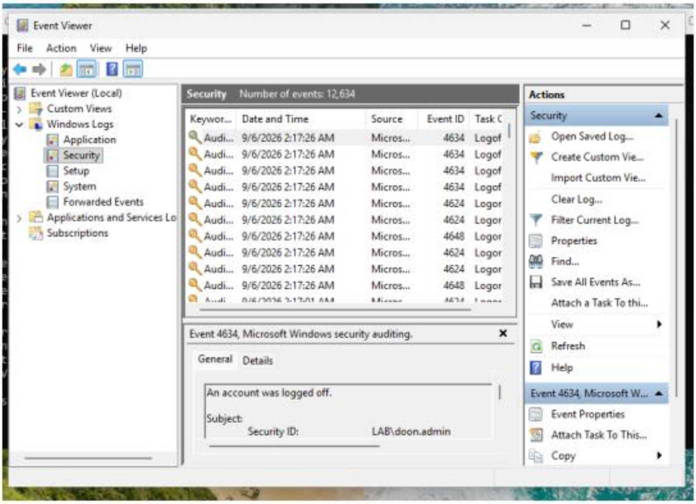


Figure 8 — Event ID 4634 for LAB\doon.admin.

A separate Event ID 4624 was observed for the CLIENT01$ computer account under SYSTEM context. It is included as evidence of successful-logon auditing without mislabeling the event as a human-user logon.

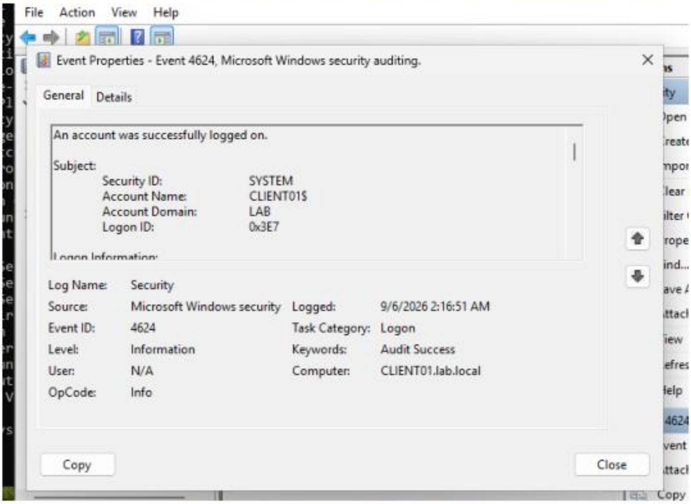


Figure 9 — Event ID 4624 for CLIENT01$ / SYSTEM.

8. Windows Defender Firewall

The Firewall Properties evidence shows the Domain Profile enabled, inbound connections blocked by default, outbound connections allowed by default, and settings controlled through Group Policy.

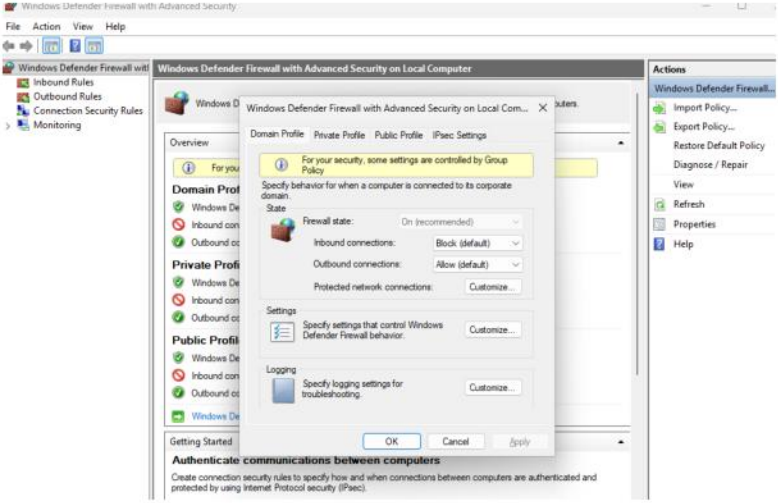


Figure 10 — Windows Defender Firewall properties.

9. Endpoint Hardening — Guest Account

The local Users view was captured during endpoint hardening. The Guest account was identified and disabled as part of the completed workstation hardening work.

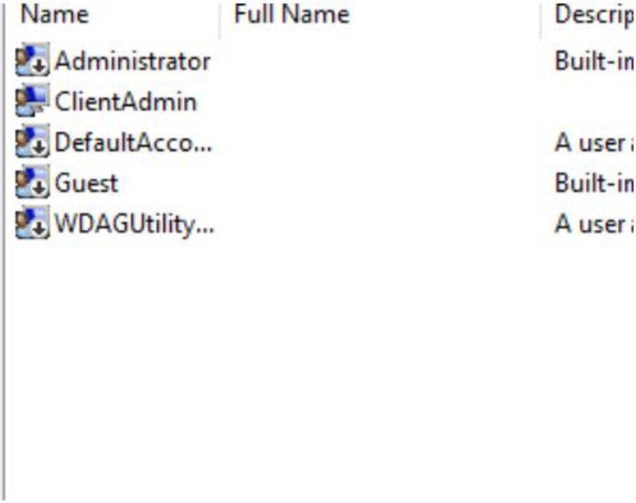


Figure 11 — CLIENT01 local Users list.

10. Troubleshooting Record

The build included an initial failed domain-join attempt caused by incorrect credentials. The failure is retained as evidence of the troubleshooting process; the domain join was subsequently completed successfully.

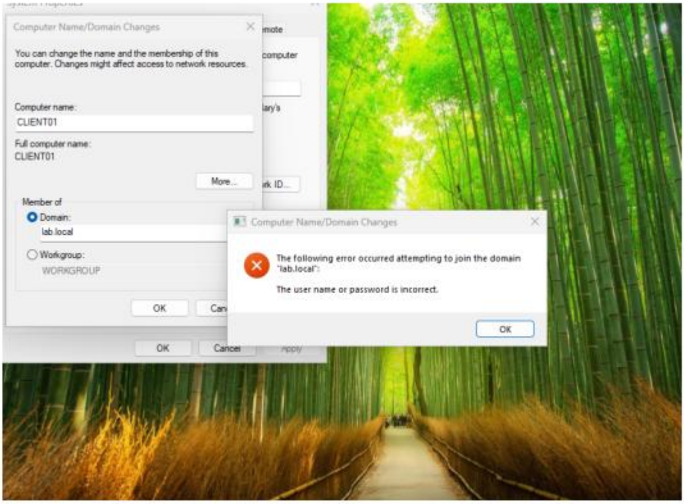


Figure 12 — Initial domain-join credential error.

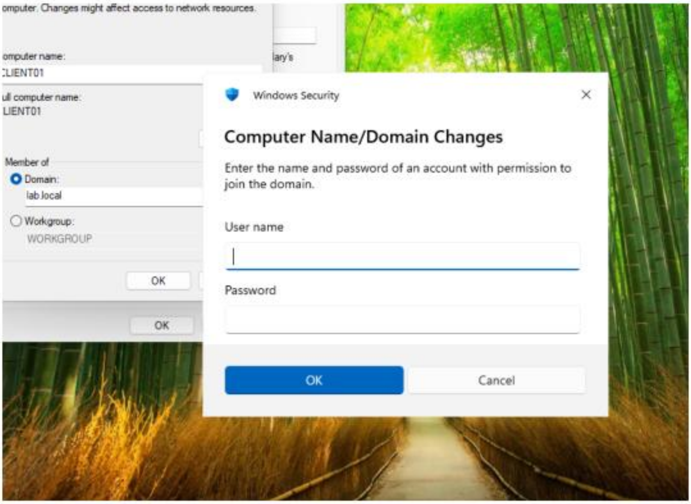


Figure 13 — Domain-join authentication prompt.

11. Final Validation & Outcome

The completed lab demonstrates a functioning Active Directory security environment with a domain controller, domain-joined workstation, structured OUs, security groups, workstation-targeted Group Policy, advanced auditing, Security event generation, Windows Defender Firewall enforcement, and endpoint hardening.

PROJECT STATUS: COMPLETE

The strongest evidence is the combination of configuration screenshots and validation screenshots: DNS resolution, ADUC OU placement, GPRESULT, auditpol, Security Event Viewer, and Firewall policy state.